Research Statement

Andrew Doppenberg

RESEARCH AREAS

My research is in **computational philosophy**, **formal social epistemology**, and the **philosophy of religion**. It also bears on Bayesian epistemology, philosophy of science, the epistemology of testimony and disagreement, modelling and idealization, and technology ethics.

Three of the oldest argument forms in philosophy of religion rest on an untested picture of how belief works. Arguments from religious disagreement hold that persistent division among sincere, capable people shows that someone must be reasoning badly. Arguments from consensus and from apostasy read public profession, or sudden waves of public doubt, as evidence about what people privately believe and how good their evidence is. Arguments from miracle testimony turn on how far multiple reports of an extraordinary event corroborate one another. All three treat believers as isolated individuals weighing fixed evidence. But belief is formed, kept, and abandoned inside families, congregations, and networks of trust, and no one has checked whether these arguments survive contact with that fact. The question cannot be settled from an armchair, because it turns on what patterns emerge when many imperfect people influence one another over time.

My research program supplies the missing tool. Agent-based modelling—computer simulation of a community of interacting individuals—forces every assumption about how people reason and trust into explicit code, varies those assumptions systematically, and shows which collective patterns follow. The method is now standard in the social epistemology of science and almost entirely absent from philosophy of religion. My dissertation built and defended it for philosophical use. The next five years apply it to religious belief.

BACKGROUND: PUBLISHED AND DOCTORAL WORK

Published work. In "Jeffrey conditionalization: against caution," *Philosophical Studies* 182 (2025): 3135–3148, I defended Jeffrey conditionalization against the objection that it can rationally lead an agent toward a false hypothesis on the basis of non-misleading observations. I used two probabilistic models to isolate the effect of the relevant observation, and then proved that the troubling cases cannot arise in general. The paper concerns how a single agent should update on evidence that never becomes certain, which is the situation of anyone who learns from testimony. The projects below carry that question from one agent to a community.

Dissertation. "Flourishing in Diversity: The Promise and Limits of Agent-Based Models" (University of Toronto) built the instrument this program applies. I argued that simulations reveal non-obvious mechanisms, expose hidden assumptions, and test whether conclusions depend on fragile idealizations, and I established principled limits on what they can show. Three case studies made the argument. The first showed that a recent skeptical result about identifying genuine experts by their track records (Huang 2024; cf. Goldman 2001) holds only under a narrow combination of assumptions. The second showed that the Zollman effect, the finding that more communication can leave a scientific community worse at reaching the truth (Zollman 2007, 2010), cannot bear the policy lessons drawn from it. The third built the first agent-based model in philosophy of religion, of Hick's soul-making response to the problem of evil (Hick 1966).

The program below grows out of this work without being a further chapter of it, and departs from it in three ways. It changes domain: existing philosophical models assume idealized inquirers exchanging cheap, honest reports, and religious communities, where testimony is bound up with authority and identity, visibly break those assumptions. It innovates methodologically: agents whose private confidence and public profession can come apart, and who pay social costs for dissent, are new to the philosophical modelling literature. And it asks a different theoretical question.

Not how a community of inquirers tracks the truth, but what an outside observer may legitimately infer from a community's visible behaviour—its disagreements, its professions, and its corroborating reports.

NEAR-TERM ARTICLES

Three articles develop dissertation chapters, and each moves past the chapter it comes from. I plan to submit all three within the coming year.

Should We Use Track Records to Identify Experts?

Status: In progress. Recent work argues that a community which defers to whoever has the best track record can do worse than one that ignores track records altogether. I show that this holds only when early records are treated as decisive, are available immediately, and rest on private evidence. Real epistemic communities rarely meet those conditions: records are noisy, recognition is delayed, and much of the evidence is public. Where the chapter established the negative result, the article adds the positive one, identifying the conditions under which track records do identify experts, and arguing that those conditions are the common case. The paper contributes to debates on expertise, deference, and the design of epistemic institutions.

The Zollman Effect and the Limits of Model-Based Policy

Status: In progress. Even granting that the standard models of transient epistemic diversity represent scientific communities accurately, a policymaker still needs answers to further questions before intervening: which measure of epistemic success matters, how much delay is acceptable, how difficult the problem is, and how much diversity is already present. The article generalizes beyond the chapter's reassessment of the Zollman literature, using that case to develop a framework for responsible policy inference from formal models of inquiry.

Modelling the Soul-Making Theodicy

Status: In progress. This article introduces agent-based modelling into the problem of evil literature and defends a conditional result: if freely helping those in need can build virtue, then an intermediate amount of suffering, a low but non-zero rate of severe suffering, and a lawlike distribution of suffering are less surprising than they first appear. The paper does not argue that actual suffering is justified. Beyond the chapter's model, it works out what the method commits a theodicist to, showing what a soul-making theodicy must assume and how much explanatory work those assumptions can do.

RESEARCH PROGRAM: A COMPUTATIONAL PHILOSOPHY OF RELIGION

The three projects below share one question: *what may an observer legitimately infer from a religious community's visible behaviour?* Each takes one visible pattern—disagreement, profession, corroboration—and asks what social conditions would produce it. The projects are independent articles, but together they treat religious belief as a socially embedded epistemic phenomenon rather than a set of individual verdicts on fixed evidence.

Religious Disagreement and Bounded Rationality

Status: Planned. Many arguments from religious disagreement assume that persistent disagreement is surprising if people are responding rationally to their evidence: were we all reasoning well, we would expect to converge (Christensen 2007; De Cruz 2019). I plan to test that assumption with agents who are limited but broadly responsible reasoners.

The model will compare tightly clustered communities with well-connected ones, equal trust with prestige-weighted trust, ideal belief revision with psychologically realistic shortcuts, evenly shared public evidence with unevenly distributed private evidence, and low with high costs of changing one's mind. My working hypothesis is that persistent disagreement is the normal condition of finite reasoners in structured social settings, and that religious diversity therefore carries little evidential weight against anyone's rationality. If instead stable disagreement requires fragile assumptions, that strengthens the case for revising one's beliefs on discovering that others disagree. Either result advances the debate by moving it from a static question, "What should I do after discovering disagreement?," to a dynamic one, "Which patterns of disagreement should we expect in different social environments?" The results bear on peer disagreement, conciliationism and steadfastness, permissivism, and the epistemology of religious diversity.

Public Profession and Private Belief

Status: Planned. Religious communities often contain a gap between private belief and public profession. People may stay publicly committed because of family pressure, livelihood, identity, or fear of scandal, even as their private confidence weakens; once enough people dissent openly, others reveal their doubts as well (Granovetter 1978; Kuran 1995). Communities can therefore look stable while being internally fragile, and a sudden wave of departures may reveal hidden doubt rather than new evidence.

I plan to model agents with private confidence, public affiliation, exit costs, social sanctions, perceptions of how many others doubt, and responses to authority scandals (Centola 2018). The model will study when slow private doubt produces no public change, when dissent cascades, and when communities restabilize after a shock, and it will separate epistemic causes of departure from social ones, since the same public pattern can arise from very different private dynamics. I expect to establish that public profession tracks private conviction only under specific conditions—low exit costs and accurate perception of others' doubts—that many real communities do not meet. This matters because arguments from religious consensus and from apostasy treat public affiliation as a guide to private belief, and private belief as a guide to the quality of the underlying evidence. If profession is sensitive to exit costs and perceived support, apparent consensus is less informative than it looks, a point with obvious parallels in the study of misinformation and political opinion (O'Connor and Weatherall 2019).

Testimonial Independence and the Argument from Miracles

Status: Planned. Bayesian treatments of miracle testimony turn, on both sides of the dispute, on a single quantity that no one has modelled. The argument that testimony can establish a miracle runs on independence: if each of n witnesses would report falsely only rarely, and each reports independently, the chance that all of them err together falls off geometrically and can outweigh any finite prior against the event (Babbage 1837; McGrew and McGrew 2009). The standard reply denies the independence, since witnesses share sources, expectations, and institutions, and their reports are copied, corrected, and preserved by communities with a stake in them (Hume 1748; Earman 2000). Both sides assert a claim about independence; neither derives it from anything.

Independence is not a brute fact about a body of testimony. It is a consequence of how that testimony was produced and transmitted, and transmission structure is exactly what an agent-based model can represent. I plan to model miracle testimony as transmission through a network and to compute, for each structure, an *effective number of independent witnesses*: how many genuinely independent reports a nominal count of n reports is worth, once shared sources, network overlap, embellishment, and institutional reinforcement are taken into account. Agents will update on testimony by Jeffrey conditionalization, so that reports raise confidence without making their content certain, which applies across a community the individual-level rule I defended in *Philosophical Studies*.

This converts a stalemate into a measurable question: how much structural independence would a testimonial tradition need for the argument from miracles to go through, and what kind of community would produce it? My working hypothesis is that effective independence saturates, rising steeply over the first few witnesses and then flattening as network overlap grows, so that large nominal witness counts are worth far less than they appear. If that is right, the argument from miracle testimony is not refuted but bounded. It can be run only for traditions with a specific structural profile—dispersed witnessing, weak common sourcing, early independent attestation—and the dispute then moves to the historical question of whether any actual tradition has that profile. That is a more tractable disagreement than the one now being had.

The project reaches past philosophy of religion. How much corroboration is worth when sources are not independent is also the problem of eyewitness testimony in law, of source criticism in history, and of rumour and misinformation, where correlated reports are routinely mistaken for converging evidence. A general treatment of effective independence would be usable in all of them.

FIVE-YEAR PLAN

The plan is ambitious but feasible, because the methodological groundwork is complete and the projects need no fieldwork, human-participant data, or ethics review.

- **Year 1:** Submit the three near-term articles. These establish the methodological base for the program: how to use models responsibly in philosophy of religion without overstating what they show.

- **Years 1–2:** Build and release the disagreement model; present at Canadian and international conferences; submit the disagreement article.

- **Years 2–3:** Build and release the profession-and-belief model; submit the second article.

- **Years 4–5:** Complete the miracle-testimony article, and write an integrative piece framing religious belief as a socially embedded epistemic phenomenon.

Each model will be released as documented open-source code with an interactive browser version, so that scholars in neighbouring fields and non-specialists can explore the results without programming. Longer term, the articles could become a book on computational philosophy of religion. Its guiding claim would be modest and, I think, more useful than any simulation result: many debates in philosophy of religion depend on assumptions about social epistemic dynamics, and those assumptions can be stated clearly, tested carefully, and argued about fruitfully once they are made explicit.

CONTRIBUTION

The program contributes at four levels. Within philosophy of religion, it brings a method still rare in the field into live debates about disagreement, consensus, and testimony, and shows which established arguments depend on unrealistic assumptions about social learning. Within social epistemology, it extends models built for scientific communities to communities organized around trust, authority, identity, and costly commitment, features the scientific-community models largely ignore. Within neighbouring empirical disciplines, the models return usable predictions: which community structures produce sudden collapses of affiliation rather than gradual decline, for instance, is a prediction sociologists of religion can set against survey data on the growth of the religiously unaffiliated. Beyond the academy, knowing when reasonable people should be expected to disagree, and when public agreement masks private doubt, matters to religious communities, secular institutions, and anyone trying to read public opinion in a period of deep division and eroding trust.

Running through all four is a methodological conviction, and the one that unifies my work: when a philosophical argument depends on a claim about how communities behave, the claim should be made explicit enough to test.